

Assignment 3: AI Pipeline

Objective

Understand how multiple AI tools, APIs, or processing steps can be connected together to automate a task.

Task

Design and build a simple **AI pipeline**.

The pipeline should contain at least **three steps**.

Example Pipeline

User Input → AI Processing → Final Output

Example:

CSV File → Python → Gemini API → Data Summary

Another example:

Article → AI Summary → Image Generation → Social Media Post

Another example:

User Question → Retrieve Information → LLM → Final Answer

Requirements

Your pipeline must include:

1. **Input**
 1. Text
 2. CSV
 3. PDF
 4. User question
 5. Image
 6. Any other suitable data
2. **Processing**

Use Python, an API, or another automation tool.
3. **AI Model**

Examples:

 1. Gemini API
 2. OpenAI API

3. Hugging Face
4. Ollama

4. Output

Examples:

1. Summary
2. Report
3. Answer
4. Generated content
5. Analysis

Pipeline Diagram

Create a simple diagram showing your workflow.

Example:

CSV File → Python → Gemini API → Analysis → Final Report

Code

Students should provide the main Python code used to build the pipeline.

Explanation

Write approximately **300–500 words** explaining:

- What problem does your pipeline solve?
- What is the input?
- What processing happens?
- Which AI model/API is used?
- What is the final output?
- What problems or limitations did you face?

Submission

Submit:

- Source code
 - Pipeline diagram
 - Sample input
 - Sample output
 - Short explanation
 - Screenshots of the working pipeline
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